Lecture 5 Live Problems

The insulin–glucose system. Feedback loops. Parameters.

Points toward lab manual pages: **IG1, IG2, IG3, IG4** (section L1 S4)

1. A mole of glucose. One mole of hydrogen atoms has mass about 1 gram. One mole of typical carbon atoms has mass about 12 grams. One mole of typical oxygen atoms has mass about 16 grams. What is the mass of one mole of glucose, $C_6H_{12}O_6$?

2. Glucose, step by step. Use the glucose change equation

$$\frac{\Delta G}{\Delta t} = m - s\, I\, G.$$

Suppose that insulin and insulin sensitivity are constants: $I = 1$ pM and $s = 0.8$. Suppose that eating and liver production yield $m = 4$. Take $\Delta t = 1$ hour for our time interval. If $G = 3$ mM now, what will G be in one hour? In two hours?

t (hours)	G (mM)	$\Delta G / \Delta t$ (mM per hour)	ΔG (mM)
0	3		
1			
2			

3. More glucose input. Same as Problem 2, but now eating and liver production yield $m = 10$. Again $I = 1$ pM, $s = 0.8$, and $\Delta t = 1$ hour. If $G = 3$ mM now, what will G be in one hour? In two hours? Compare with Problem 2.

t (hours)	G (mM)	$\Delta G / \Delta t$ (mM per hour)	ΔG (mM)
0	3		
1			
2			